CS5224: Cloud Computing

Preliminary Report (Group 28 - TrolleySaver)

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Motivation

Problem Statement

A household grocery planner living in an HDB estate does a weekly shop for staples, such as rice, cooking oil, milk, eggs, bread and other household items. Three or four supermarket branches are close enough to visit. Each uses different shelf prices, pack sizes and promotions and some offers are only limited to members or to shoppers who buy in bulk. Before shopping, the planner would want to know the following:

1. Is the same item cheaper at another branch?
2. Which single store is the cheapest for the whole grocery list?
3. Is there a healthier option in this category that does not cost much more than the original?

Today, answering these questions means visiting stores, checking apps or websites one retailer at a time, comparing pack sizes by hand and working out whether offer conditions apply. Most shoppers do not have the time for this every week, going back to their usual store to purchase their groceries, which may or may not be the cheapest option.

Pain Points

The pain points listed below are the working hypothesis for this SaaS product:

- **Prices are scattered across retailers:** The same product can cost a different amount at different retailers and branches. Checking means visiting several stores or looking up each retailer separately.
- **The whole basket is hard to optimise:** The cheapest store for one item may not be the cheapest for the full list. Totalling a full list across several stores takes too much effort and time to do by hand.
- **Affordable healthier options are hard to find:** Healthier choice products exist in many categories, but there is no easy way to compare their prices with other alternatives. Finding healthier choice products and checking prices are two separate tasks.

Business Idea

Why we are solving this problem

Singapore's food inflation rose to 2.2% year-on-year in July 2026, up from 2.1% in June 2026. The Monetary Authority of Singapore (MAS) and the Ministry of Trade and Industry (MTI) attributed the increase to faster price increases for both food services and non-cooked food (MAS, 2026). Non-cooked food is the part of this measure closest to grocery shopping, so households buying groceries are directly affected. Over the same period, headline inflation rose from 1.9% to 2.2% (Department of Statistics, 2026).

In a CASE poll of 366 Price Kaki app users after Budget 2026, 59.5% said grocery costs had risen the most over the past 12 months. Another 58.5% said keeping prices of daily essentials stable would give them the greatest relief (Lim, 2026).

In 2023, resident households in Singapore spent an average of $455.90 per month on food and non-alcoholic beverages according to SingStat's Household Expenditure Survey (HES), which is carried out every 5 years. (SINGSTAT, 2024). Because this spending happens every month, even a small percentage saving on a household's grocery basket can add up over a year.

Together, these points show that grocery costs are indeed rising, with shoppers feeling the increase and groceries make up a meaningful share of monthly household expenditure. Shoppers cannot control the prices of essential items, but they can choose where and what to buy. Our SaaS app aims to make that choice easier by reducing the effort needed to compare prices across nearby stores.

Existing Solutions

Grocery price comparison applications already exist in Singapore. The two most relevant services are GroceryHunt and Price Kaki by the Consumers Association of Singapore (CASE).

- **GroceryHunt:** Compares prices across Sheng Siong, FairPrice, and Cold Storage with deals, unit pricing, price history, and product search (updated daily). However, it does not describe branch-level prices, whole-basket planning, Healthier Choice Symbol (HCS) integration, or how prices are collected.
- **Price Kaki:** A community-driven price comparison service for supermarkets and hawker centres where users filter by price, distance, and retailer, receive price-change alerts, and earn rewards for updates. However, it lacks a planner to calculate the cheapest total for a full shopping list or evaluate if a second stop is worth the travel.

Our Solution

TrolleySaver will be a cloud-hosted web application that serves two distinct user groups: Shoppers who utilize price-comparison and basket optimization features, and Contributors who supply community price data. Both groups can use the app in a browser on a phone or computer, with no installation needed. It has the following three features:

1. Collect prices. Contributors submitted prices observed at specific branches. Each submission records the product, pack size, branch, time observed, price type (whether it's regular, promotion, or member price) and any other offer condition.
2. Plan the basket. The shoppers enter a list and their preferences, including an option to choose only Healthier Choice Symbol products. The system compares every allowed single store.
3. Pick the cheapest option. Shoppers will give us their shopping list and the system tells them which single store gives the lowest overall price for their whole list.

Some of the major technical risk include

1. **Sparse or incomplete price data:** In the early stages, contributor coverage will be uneven. Hence, many prices would not be documented. A potential solution is to mitigate the problem is by flagging missing price items as "no price found, working on it" and maybe implement a reward scheme for members that contribute prices for high demand missing price items
2. **Unreliable or inconsistent contributor submissions:** Prices may be mistyped,or logged against the wrong branch or pack size, undermining trust in the recommendations.A potential solution is to constrain submissions to dropdowns from a fixed product catalog and branch list instead of free text.

Expected Prototype Scope:

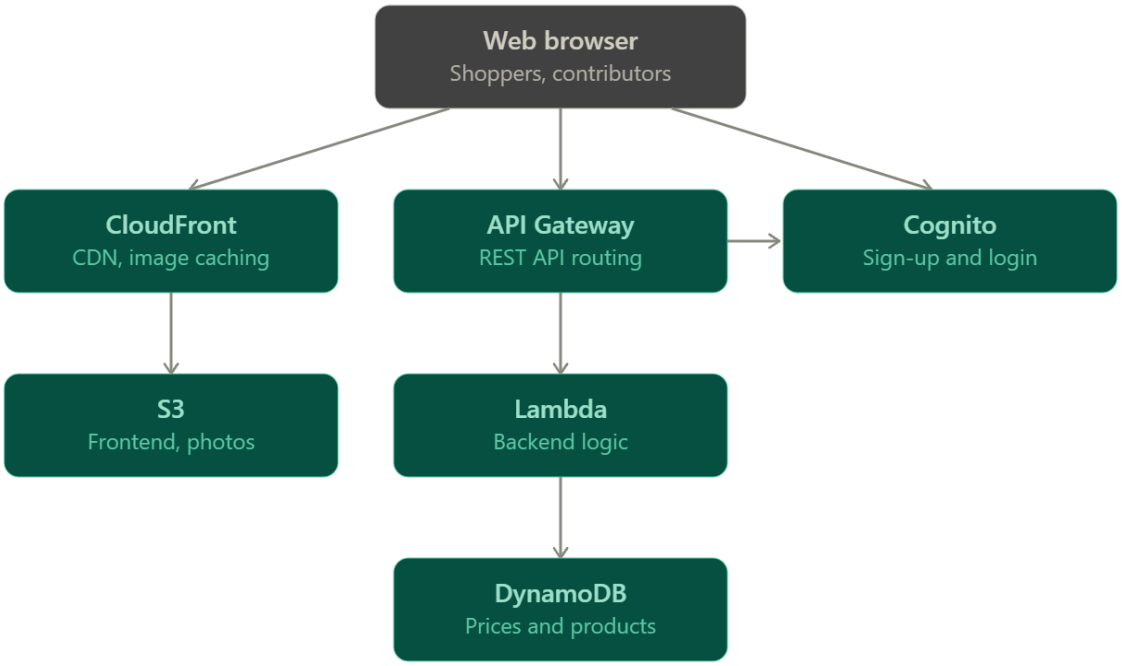
1. 2–3 pilot supermarket branches
2. A fixed catalog of 10-20 common grocery items.
3. Contributor price submissions: product, pack size, branch, time observed etc.
4. Basket planning across single stores and all store pairs within the pilot set
5. Basic Healthier Choice filter to identify healthier products at similar prices
6. System and user evaluation with a small group of testers'
7. Proof of Concept ready for final presentation

<u>Business Model</u>

TrolleySaver will be free for shoppers. The main value is received by shoppers, who can save time and potentially reduce their grocery expenditure. Users can compare grocery prices across supermarkets, optimise their grocery basket, and identify affordable healthier alternatives. A free model reduces the barrier to adoption and allows the service to build a larger user base and price database.

The service will explore potential revenue if the platform reaches sufficient scale. If the service grows beyond the prototype stage, potential revenue could come from business subscriptions and partnerships. Supermarkets and consumer-goods companies could pay for aggregated price trends and market insights, while retailers or brands could pay for clearly labelled sponsored promotions. The core consumer price-comparison service would remain free.

Preliminary Architecture & Design



Functional Requirements

- Contributors can submit prices with all required fields and an optional photo.
- Shoppers can search products and enter a basket with preferences.
- The system recommends the lowest-cost single-store. The system produces a per-store shopping list with quantities, subtotals, and unpriced items.
- Users can register and log in, and every submission is tied to a user.

Price Validation & Selection Logic

- Each submission records product, pack size, retailer, time observed, price type (regular, promotion or member) and any offer condition, with an optional shelf-tag photo.
- On submission, check required fields and a valid price type, reject duplicates, and flag prices outside a plausible range based on that item's price history at the retailer.
- Mark a price stale once it exceeds an age limit, shorter for promotions, with a contributor-supplied offer end date taking precedence.
- For each item at each retailer, use the latest non-stale price the shopper is eligible for: member prices only if the shopper holds that membership, and conditions such as "buy 2" only if the requested quantity meets them.
- Return items with no eligible price as unpriced rather than omitting them.

Basket Optimisation

- Total the basket at every allowed retailer using the shopper's eligible prices.
- Recommend the single retailer with the lowest total, and show the per-item breakdown alongside any unpriced items.

Assumptions

- All users are in one region, so one AWS region (ap-southeast) is enough.
- S3 buckets are private and only accessible via CloudFront or presigned URLs.
- Limit upload size and file type, and rate-limit submissions per user for abuse prevention.

Cloud Services Used

The service selection below is presented using AWS terminology, but the architecture is not tied to any single provider. AWS was selected for the prototype primarily for its free tier coverage and the team's familiarity with its tooling.

Service Type	Service	Purpose in the System
Object Storage	**Amazon S3**	Stores user-uploaded product and price-tag photos submitted as evidence alongside price reports, plus hosts the static frontend build.
NoSQL Database	**Amazon DynamoDB**	Stores products, price submissions, store locations, user profiles, and watchlists. Price submissions are keyed by product ID so all reports for an item are retrieved in a single query. A secondary index supports lookups by store.
Identity & Access Management	**Amazon Cognito**	Handles user registration and authentication. User identity is tied to every price submission, enabling reputation tracking and abuse prevention.
Serverless Compute	**AWS Lambda**	Backend logic for product search, basket optimization, price submission handling, confidence-score recalculation, and stale-price flagging.
API Layer	**Amazon API Gateway**	Exposes the Lambda backend as a REST API to the web frontend, handling routing, request validation, and Cognito-based authorization.
CDN	**Amazon CloudFront**	Serves the static frontend and caches product images globally, reducing repeated S3 requests.

Implementation Plan

Development phases and timeline for the next 8 weeks before the final presentation is shown in the table below:

Phase	Timeline	Key activities
1. Define Scopes and Requirements	Sep 20 - Sep 28	1. Define business idea, business model, potential cloud services used. 2. Craft preliminary evaluation plan and architecture models. 3. Write preliminary report
2.Design and Data Collection	Sep 29 - Oct 5	1. Create the app's initial UI/UX design 2.Develop backend database schema to accommodate data such as supermarket locations, prices of each product, shopping list submissions etc. 3. Finetune system and cloud architecture 4. Collect prices of everyday goods from various supermarket chains
3. App Feature: Community Led Price and Product Collection	Oct 6 - Oct 12	1. Create price collection form frontend 2. Integrate with backend database 3. Basic form validation to prevent repeat or garbage inputs.
4.App Feature: Smart Basket	Oct 13 - Oct 26	1. Develop frontend to collect user's weekly shopping list. 2.Develop backend comparison logic across single stores to achieve the lowest cost.
5.App Feature: "Healthier Choice" filter	Oct 27 - Nov 1	1.Develop a filter frontend that highlight healthier alternatives at similar prices using HPB's Healthier Choice Symbol data
6.Testing	Nov 2 - Nov 8	1.Conduct user and system testing as documented below
7.Pilot and Wrap Up	Nov 9 - Nov 13	1.Final fixes from testing 2. Write final report and prepare for final presentation

Responsibilities of each group member

Role	Team Member	Key Responsibilities

Backend Developer 1 : Data & Submissions	Yong Hsien	1. Design and implement database to store community price submissions
Backend Developer 2 : Optimization Engineer	Gareth	1. Implement store comparison logic, healthier choice backend filter logic, lowest-cost calculation.
Frontend Developer	Nancy	1. Create app wireframes 2. Develop List entry and preferences form, results/plan display, shopping list output 3. Develop price submission form for community contributors
Cloud Engineer	Xinpeng	1.Select and provision the cloud provider/services (compute, database, storage) 2.Set up dev/staging/production environments
Quality Assurance Engineer	Meng Zhun	1. Conducts stress testing and bug tracking 2. perform system and user evaluation
Project Lead	Tien You	1. Create scope and timeline, ensure the team is on track 2. Assist other team members when needed

Evaluation Plan

As a Proof-of-Concept, prices are collected manually and the goal is to demonstrate the service's functionality rather than a real-time market coverage.

System evaluation

- Smart Basket recommendation will be tested by comparing the system's recommended retailer and basket total against manually calculated totals for each retailer. We will be measuring the savings achieved relative to purchasing the complete basket from each individual supermarket.
- Search quality is evaluated using alternative product queries (abbreviations, capitalisations and typos) to determine if the intended product will return.
- Image-upload function will be tested using proper image upload and ensure that the image is properly rendered and stored in our database.
- Healthier choice filter will be compared against manually verified HPB labels.

User evaluation

Test users will go through a shopping journey: compare a product, create shopping baskets, identify cheapest shopping location, locate healthier options and upload product images. They will have to compare themselves completing the task manually using separate supermarket websites. We will use metrics such as accuracy, completion time and price to determine the success. Satisfactory surveys will also be used to rate the usefulness and ease of use of the product. The team will also conduct an on-demand interview to obtain more qualitative feedback.

References

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